1 Problem and data description

1.1 Task and research question

The task is supervised image classification: assign one clothing category to each image. Category prediction can help organize online product catalogs. The study asks how linear, multilayer, convolutional, recurrent, and Transformer models differ in accuracy and class confusions on the same data splits.

For Fashion-MNIST, the model receives a normalized grayscale tensor $x \in \mathbb{R}^{1\times28\times28}$ and returns ten class scores $f_\theta(x) \in \mathbb{R}^{10}$. Its prediction is $\hat{y} = \arg\max_{k\in\{0,\dots,9\}} f_\theta(x)_k$. The prediction unit is one image. Low resolution and similar garment outlines make some categories difficult to separate.

1.2 Dataset and study scope

Fashion-MNIST contains 70,000 Zalando product images with supplied category labels: 60,000 official training images and 10,000 official test images [3]. Labels 0–9 denote T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot, respectively.

We use the standard IDX distribution from the authors' MIT-licensed repository [4]. The four cached gzip files total 30.88 MB and match the published MD5 checksums. Images are unsigned 8-bit arrays; labels are integer class IDs. No samples are removed or relabelled. The small, isolated product images and fixed category set limit how far results extend to full scenes or unfamiliar products.

Fashion-MNIST is the main comparison dataset. MNIST handwritten digits are used for development, and CIFAR-10 colour images provide a separate extension [5, 6]. Their roles and working splits appear in Table 1.

Table 1: Dataset roles, native image sizes, and split counts. Image dimensions are height × width × channels. Counts refer to images.

Dataset	Role	Image size	Train	Validation	Test
Fashion-MNIST	Main comparison	$28 \times 28 \times 1$	54,000	6,000	10,000
MNIST	Development	$28 \times 28 \times 1$	54,000	6,000	10,000
CIFAR-10	Extension	$32 \times 32 \times 3$	45,000	5,000	10,000

1.3 Exploratory data analysis

The following statistics use the full, unaugmented official Fashion-MNIST training set, before the train/validation split. Every image is 28×28 pixels. Each class has 6,000 images, so the largest-to-smallest class ratio is 1.00. The official test set has 1,000 images per class. Figure 1 shows one real example per class.

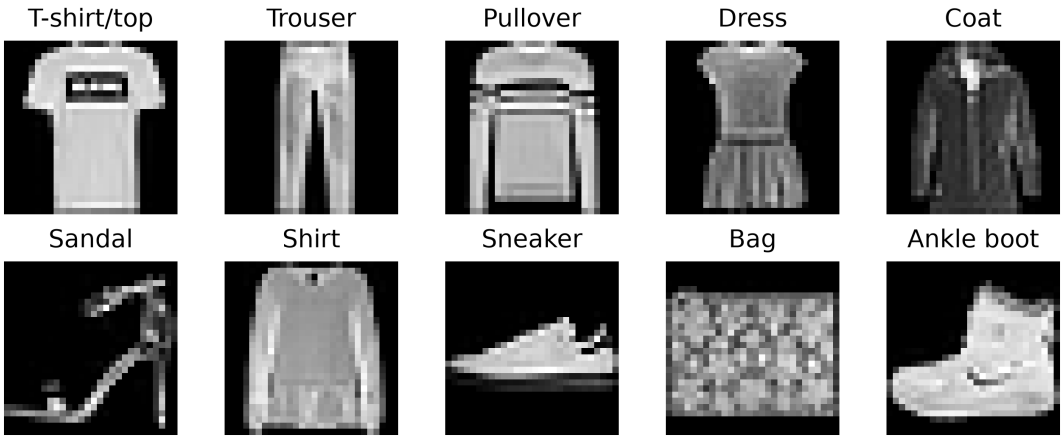


Figure 1: Fashion-MNIST: the first image of each class in official training order, selected without using model predictions.

Pixel intensity has mean 72.94 and standard deviation 90.02 on the 0–255 scale. About 53.0% of pixels fall in the lowest histogram bin (intensities 0–7), reflecting the dark background. Defining foreground as intensity above 32 gives average coverage from 24.5% for Sandal to 59.8% for Pullover.

Figure 2 compares class-average images after subtracting the average across classes. Cosine similarity is 0.86 for Pullover–Coat and 0.82 for both Sandal–Sneaker and Pullover–Shirt. These pairs motivate later inspection of class confusions; prototype similarity does not measure model errors.

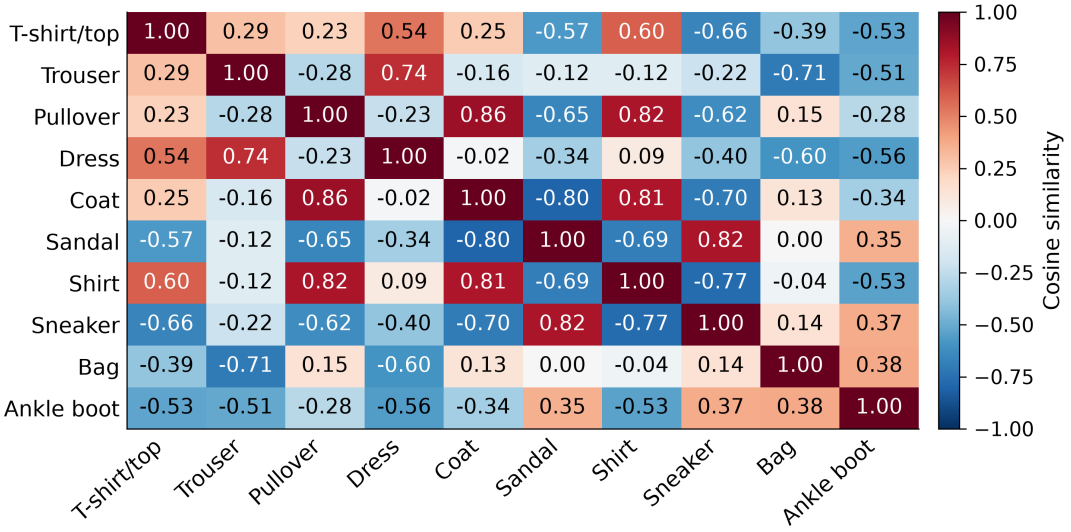


Figure 2: Cosine similarity between centred Fashion-MNIST class-average images, computed from the 60,000 official training images.

1.4 Splits and leakage prevention

For each dataset, a random permutation with seed 42 reserves 10% of the official training set for validation. The split unit is one image, and sampling is not stratified. For Fashion-MNIST, class counts therefore range from 5,353 to 5,445 in training and from 555 to 647 in validation. All models use the same indices for a given dataset.

The official test sets remain separate from training and validation. Validation loss selects checkpoints; test results are reserved for reporting. Training and validation indices are disjoint. No additional duplicate-image or product-level audit is recorded, so this separation does not establish independence between all images.

1.5 Preprocessing and augmentation

MNIST and Fashion-MNIST keep their native resolution. Pixels are converted to float32 in $[0, 1]$, then normalized as $x' = (x - 0.5)/0.5$. Training adds random rotation within $\pm 10^\circ$ and translation up to 10% of image size.

CIFAR-10 uses four-pixel padding, a 32×32 random crop, and a random horizontal flip for training. We normalize its RGB channels using mean $(0.4914, 0.4822, 0.4465)$ and standard deviation $(0.2470, 0.2435, 0.2616)$. Validation and test data receive conversion and normalization only.

An inspected Fashion-MNIST validation batch has float32 image shape $[128, 1, 28, 28]$ and values from -1 to 1. Its int64 labels have shape $[128]$; the first eight IDs are $[9, 2, 5, 0, 5, 1, 3, 5]$. In the shared $[B, C, H, W]$ convention, B is batch size, C is channels, and H, W are image height and width. CIFAR-10 batches use shape $[B, 3, 32, 32]$.

References

[3] Han Xiao, Kashif Rasul, and Roland Vollgraf, *Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms*, arXiv:1708.07747, 2017. https://arxiv.org/abs/1708.07747.

[4] Zalando Research, *Fashion-MNIST dataset and distribution files*. https://github.com/zalandoresearch/fashion-mnist. Published checksums, label mapping, and repository MIT license; accessed 27 October 2026.

[5] PyTorch contributors, *torchvision.datasets.MNIST documentation*. Online documentation. Accessed 27 October 2026.

[6] Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton, *CIFAR-10 and CIFAR-100 datasets*. https://www.cs.toronto.edu/~kriz/cifar.html. Accessed 27 October 2026.